

Electric Circuits II Project

Report

Prepared by

Name	ID
Student 1	-----
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Tutorial Number

Theoretical background:

Provide an explanation about the chosen filter and the related equations for the gain, and half-power frequencies (or cut-off frequency).

Filter design:

Provide the values for the components and parameters (Resistance/Capacitance values, half power frequencies, ..., etc.) to be used in the simulation and hardware implementation.

Circuit simulation:

Includes the detailed steps for the circuit construction on the software, i.e. screenshots for the schematic diagram and simulation profiles.

NB: You can refer to any lab manual which included simulations on PSpice for aid.

Simulation results:

Provide the screenshots for the obtained simulation results and comments on the simulation results.

You need to run two simulation profiles:

i. AC sweep simulation:

to sketch the gain versus the frequency, where V_{ac} is used as the input signal (One sketch is required).

NB: when you acquire the results from the AC Sweep, you should make sure the gain is ≥ 1

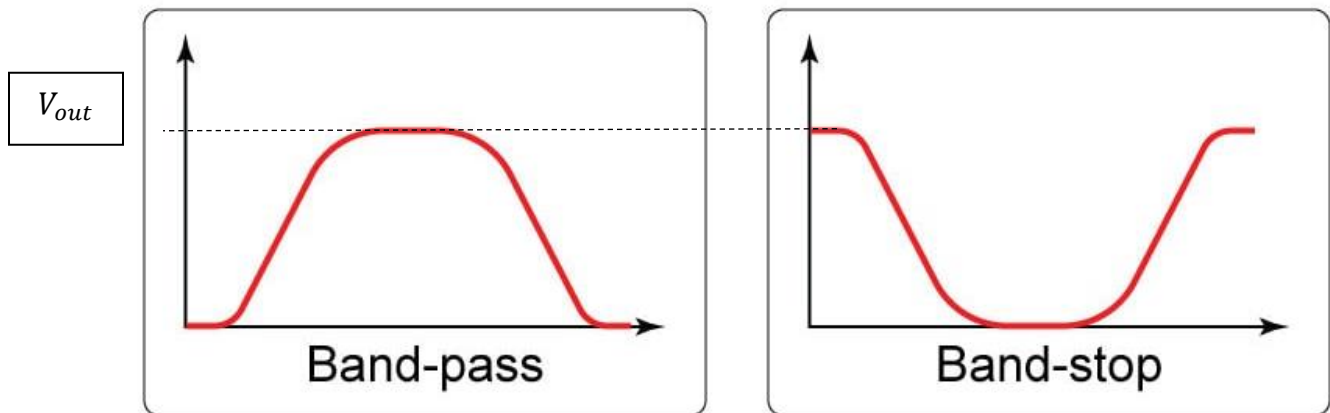


Figure 1 Expected outcome for the AC Sweep

ii. Time domain simulation:

to sketch voltage amplitude versus time, where V_{sin} is used as the input signal at three different frequencies ($f < f_o$, $f = f_o$ and $f > f_o$) (three sketches are required).

Hardware implementation:

This should include the details of the purchased components and pictures for the hardware implementation of the filter circuit.

Hardware results:

Includes pictures from the oscilloscope for the input voltage and the output **voltage at three different frequencies** (to show that the type of filter).

Includes also discussion of the results obtained from the hardware implementation.

i. $f < f_0$:

ii. $f = f_0$:

iii. $f > f_0$: